

HADOOP COMPONENTS

1. Pig

Apache Pig is a platform for analysing large datasets that consists of:

- **A high-level data flow language:** Pig Latin
- **An execution framework:** that translates scripts into a series of MapReduce jobs

Pig was developed at Yahoo! to overcome the limitations and complexities of writing raw MapReduce programs. It offers a more intuitive approach to data processing.

Key Features

- Simplifies writing complex MapReduce jobs.
- Handles both **structured and semi-structured** data.
- Supports **ETL (Extract, Transform, Load)** tasks effectively.
- Operates in both **local** and **Hadoop distributed modes**.
- Can extend functionality through **User Defined Functions (UDFs)** in Java, Python, etc.

Pig Architecture

Apache Pig's architecture consists of two main components:

=> Pig Latin Language

Pig Latin is the scripting language used to express data transformations. It is:

- **Procedural** in nature (sequence of steps)
- Easy to learn and write
- Closer to SQL, but designed for parallel dataflow execution

=> Pig Execution Engine

Responsible for executing Pig Latin scripts. It has two modes:

- **Local Mode:** Executes scripts on a single JVM (no Hadoop)
- **MapReduce Mode:** Scripts are converted into MapReduce jobs and executed on Hadoop clusters

Execution Mechanism

Pig converts the Pig Latin script into a **logical plan**, which is optimized and transformed into a **physical plan** consisting of MapReduce jobs.